M3503 WEEK 9 2022

POWER SERIES

A power series in $(x-a)$ or a power series centered at a is an infinite series of the form

$$\sum_{n=0}^{\infty} C_n (x-x_0)^n = C_0 + C_1(x-a) + C_2(x-a)^2 + \cdots$$

IDENTITY PROPERTY

If $\sum_{n=0}^{\infty} C_n (x-x_0)^n = 0$ for all x in some open interval, then

$C_n = 0$ for all n.

SHIFTING THE SUMMATION INDEX

The index of summation is just a variable, that is,

$$\sum_{n=0}^{\infty} C_n (x-x_0)^n = \sum_{k=0}^{\infty} C_k (x-x_0)^k \quad \text{if we take } n=k.$$

Example

Write $\sum_{n=3}^{\infty} n^2 a_{n-1} (x+4)^{n+2}$ as a series that starts at $n=0$

Soln

Let $k = n-3$ so that when $n=3$, $k=0$

$\Rightarrow n = k+3$. Substituting into the series, we get

$$\sum_{k=0}^{\infty} (k+3)^2 a_{k+2} (x+4)^{k+5}$$

$$\therefore \sum_{n=3}^{\infty} n^2 a_{n-1} (x+4)^{n+2} = \sum_{n=0}^{\infty} (n+3)^2 a_{n+2} (x+4)^{n+5}$$

② Express the series

$$\sum_{n=1}^{\infty} n C_n X^{n-1}$$

as series with generic term X^n instead of X^{n-1}

Soln

let $K = n-1 \quad \Rightarrow n = K+1$

$$\sum_{K=0}^{\infty} (K+1) C_{K+1} X^K$$

$$\therefore \sum_{n=1}^{\infty} n C_n X^{n-1} = \sum_{n=0}^{\infty} (n+1) C_{n+1} X^n$$

Addition Of Power Series

In order to add two or more series given in summation notation, it is necessary that the series indices start with the same number and that the powers of x are in phase, that is, if one series starts with X^p, then this should be the case with all the other series.

Examples

Writh the expression as a single series whose general term involves X^n

① $\sum_{n=1}^{\infty} n C_n X^{n-1} - \sum_{n=0}^{\infty} C_n X^n$

② $\sum_{n=2}^{\infty} n(n-1) C_n X^{n-2} - 2\sum_{n=1}^{\infty} n C_n X^n + \sum_{n=0}^{\infty} C_n X^n$

Soln

① $\sum_{n=1}^{\infty} n C_n x^{n-1} - \sum_{n=0}^{\infty} C_n x^n$

Note that the powers of x are in phase since the starting power of x in both series is x^0. Next we make the starting index to be the same by letting $K = n-1$ in the first series

$$\sum_{n=1}^{\infty} n C_n x^{n-1} = \sum_{K=0}^{\infty} (K+1) C_{K+1} x^K = \sum_{n=0}^{\infty} (n+1) C_{n+1} x^n, \text{ so}$$

$$\sum_{n=1}^{\infty} n C_n x^{n-1} - \sum_{n=0}^{\infty} C_n x^n = \sum_{n=0}^{\infty} (n+1) C_{n+1} x^n - \sum_{n=0}^{\infty} C_n x^n$$

$$= \sum_{n=0}^{\infty} [(n+1) C_{n+1} - C_n] x^n$$

② $\sum_{n=2}^{\infty} n(n-1) C_n x^{n-2} - 2 \sum_{n=1}^{\infty} n C_n x^n + \sum_{n=0}^{\infty} C_n x^n$

Powers of x are not in phase; 1st series starts with x^0, 2nd series starts with x^1 and 3rd series start with x^0. To have powers of x in phase, we write out the first terms of the 1st and the third series.

$$2C_2 + \sum_{n=3}^{\infty} n(n-1) C_n x^{n-2} - 2 \sum_{n=1}^{\infty} n C_n x^n + C_0 + \sum_{n=1}^{\infty} C_n x^n$$

To make the index 1 in the 1st series, we let

$K = n-2$, so

$$\sum_{n=3}^{\infty} n(n-1) C_n x^{n-2} = \sum_{K=1}^{\infty} (K+2)(K+1) C_{K+2} x^K = \sum_{n=1}^{\infty} (n+2)(n+1) C_{n+2} x^n$$

Thus we have

$$C_0 + 2C_2 + \sum_{n=1}^{\infty} (n+2)(n+1) C_{n+1} x^n - 2 \sum_{n=1}^{\infty} n C_n x^n + \sum_{n=1}^{\infty} C_n x^n$$

$$= C_0 + 2C_2 + \sum_{n=1}^{\infty} [(n+2)(n+1)C_{n+1} - 2nC_n + C_n]x^n$$

$$= C_0 + 2C_2 + \sum_{n=1}^{\infty} [(n+2)(n+1)C_{n+1} - (2n-1)C_n]x^n$$

Analytic Functions

A function f is analytic at x_0 if f can be expressed as a power series around x_0, with positive or infinite radius of convergence.

A polynomial is analytic at every point. This is because a polynomial is infinitely differentiable, thus we can always find its Taylor series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n$$

about x_0, which is a special kind of power series.

A rational function $\frac{P(x)}{Q(x)}$, where $P(x)$ and $Q(x)$ are polynomials without a common factor, is analytic except at x_0 for which $Q(x_0) = 0$

Recall Maclaurin series representation of f

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$$

FUNCTION	MACLAURIN SERIES
e^x	$\sum_{n=0}^{\infty} \frac{1}{n!} x^n = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$
$\cos(x)$	$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$
$\sin(x)$	$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$
$\tan^{-1}(x)$	$\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} x^{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots$
$\cosh(x)$	$\sum_{n=0}^{\infty} \frac{1}{(2n)!} x^{2n} = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \cdots$
$\sinh(x)$	$\sum_{n=0}^{\infty} \frac{1}{(2n+1)!} x^{2n+1} = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots$
$\ln(x+1)$	$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$
$\frac{1}{1-x}$	$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots$

Power Series Solutions To Linear Differential Equations

Example

Find power series solution $y = \sum_{n=0}^{\infty} C_n x^n$ of

$y' - 5y = 0$

Soln

Substituting y into the equation, we get

$$\sum_{n=1}^{\infty} n C_n x^{n-1} - 5 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} (n+1) C_{n+1} x^n - 5 \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\Rightarrow \sum_{n=0}^{\infty} [(n+1) C_{n+1} - 5 C_n] x^n = 0$$

Applying the identity property or comparing coefficients of x^n, we get

$$(n+1) C_{n+1} - 5 C_n = 0$$

$$\Rightarrow C_{n+1} = \frac{5 C_n}{n+1}, \quad n = 0, 1, 2, \ldots$$

for $n=0$: $C_1 = 5 C_0$

$$n=1 : C_2 = \frac{5 C_1}{2} = \frac{5^2 C_0}{2}$$

$$n=2 : C_3 = \frac{5 C_2}{3} = \frac{5^3 C_0}{3 \cdot 2}$$

$$n=3 : C_4 = \frac{5 C_3}{4} = \frac{5^4 C_0}{4 \cdot 3 \cdot 2}$$

$$y = \sum_{n=0}^{\infty} C_n x^n = C_0 + C_1x + C_2x^2 + C_3x^3 + C_4x^4 + \cdots$$
$$= C_0\left[1 + 5x + \frac{5^2}{2}x^2 + \frac{5^3}{3!}x^3 + \frac{5^4}{4!}x^4 + \cdots\right]$$
$$= C_0 \sum_{n=0}^{\infty} \frac{(5x)^n}{n!}$$

Power Series Solution About Ordinary Points

Consider a second order homogeneous linear equation

$$a_2(x)\,y'' + a_1(x)\,y' + a_0(x)\,y = 0 \quad \text{— ①}$$

Dividing by $a_2(x)$, we get

$$y'' + P(x)\,y' + Q(x)\,y = 0 \quad \text{— ②}$$

x_0 is an ordinary point of equation ① if both $P(x)$ and $Q(x)$ in equation ② are analytic at x_0, otherwise x_0 is a singular point of equation ①.

Examples

Determine singular points of

① $(x^2-2)y'' + 2y' + \sin(x)\,y = 0$

② $(x^2-x-2)y'' + (x+1)y' - (x-2)y = 0$

③ $\sin(x)\,y'' + \cos(x)\,y = 0$

④ $\sin(x)\,y'' - \ln(x)\,y = 0$

Soln

① $x^2 - 2 = 0$

$\Rightarrow x = \pm\sqrt{2}$ //

② $(x-2)(x+1) = 0$

$\Rightarrow x = 2,\ x = -1$ //

③ $\sin(x) = 0$

$\Rightarrow \sin^{-1}(0) = x$

$\Rightarrow x = n\pi$, n is an integer

④ $\ln(x)$ is undefined for $x \leq 0$, and

$\sin(x) = 0 \Rightarrow x = n\pi$, $n = 0, \pm 1, \pm 2, \ldots$

$\therefore x \leq 0$ and $x = n\pi$, $n = 1, 2, 3, \ldots$ are singular points

Theorem : Existence of power series solution.

If x_0 is an ordinary point of equation ① , we can always find two linearly independent solutions in the form of a power series centered at x_0.

The radius of convergence of any power series solution of the form $\sum_{n=0}^{\infty} C_n (x - x_0)^n$ is at least as big as the distance R from x_0 to the nearest singular point (real or complex valued) of equation ①. The distance R is also known as the minimum radius of convergence.

Examples

Find the minimum radius of convergence R of

① $(x+1)y'' - 3xy' + 2y = 0$ about $x_0 = 1$

② $y'' - \tan(x)y' + y = 0$ about $x_0 = 0$

③ $(x^2 - 2x + 5)y'' + xy' - y = 0$ about $x = -1$

Soln

① $x+1=0 \Rightarrow x=-1$

$R=\sqrt{(1+1)^2}=2$

② $\cos x=0 \Rightarrow x=\cos^{-1}(0)$

$=\frac{\pi}{2}+n\pi$, n is an integer

The distance between 0 and $\pi/2+n\pi$ is

$\pi/2+n\pi$, $n=0,1,2,3,\ldots$

which is minimum when $n=0$

$\therefore$ Minimum radius of convergence is $\pi/2$

③ $x^2-2x+5=0$

$\Rightarrow x=\frac{2\pm\sqrt{4-4(5)}}{2}$

$=1\pm 2i$ recall $-1=-1+0i$

$R=\sqrt{(-1-1)^2+(0-2)^2}=\sqrt{8}=2\sqrt{2}$

Find power series solution of the given equation about ordinary point $x=0$

① $y''-(x+1)y'-y=0$

② $(x-1)y''-xy'+y=0$; $y(0)=-2$, $y'(0)=6$

③ $y''+\sin(x)y=0$

Soln

① Substituting $y=\sum_{n=0}^{\infty}C_nx^n$ gives

$$\sum_{n=2}^{\infty}n(n-1)C_nx^{n-2}-(x+1)\sum_{n=1}^{\infty}nC_nx^{n-1}-\sum_{n=0}^{\infty}C_nx^n=0$$

$$\Rightarrow \sum_{n=2}^{\infty}n(n-1)C_nx^{n-2}-\sum_{n=1}^{\infty}nC_nx^n-\sum_{n=1}^{\infty}nC_nx^{n-1}-\sum_{n=0}^{\infty}C_nx^n=0$$

$$\Rightarrow 2C_2 + \sum_{n=3}^{\infty} n(n-1)C_n x^{n-2} - \sum_{n=1}^{\infty} nC_n x^n - C_1 - \sum_{n=2}^{\infty} nC_n x^{n-1} - C_0 - \sum_{n=1} C_n x^n = 0$$

$$\Rightarrow 2C_2 - C_1 - C_0 + \sum_{n=1}^{\infty} (n+2)(n+1)C_{n+2}x^n - \sum_{n=1}^{\infty} nC_n x^n - \sum_{n=1}^{\infty} (n+1)C_{n+1}x^n - \sum_{n=1}^{\infty} C_n x^n = 0$$

Comparing coefficients, we get

x^0 : $2C_2 - C_1 - C_0 = 0$

$\Rightarrow C_2 = \frac{1}{2}C_0 + \frac{1}{2}C_1$

x^n : $(n+2)(n+1)C_{n+2} - nC_n - (n+1)C_{n+1} - C_n = 0$

$$\Rightarrow C_{n+2} = \frac{(n+1)C_{n+1} + (n+1)C_n}{(n+2)(n+1)}$$

$$= \frac{C_{n+1} + C_n}{n+2}, \quad n = 1, 2, 3, \cdots$$

For $n=1$, $C_3 = \frac{C_2 + C_1}{3} = \frac{\frac{1}{2}C_0 + \frac{1}{2}C_1 + C_1}{3}$

$$= \frac{C_0}{6} + \frac{C_1}{2}$$

For $n=2$, $C_4 = \frac{C_3 + C_2}{4} = \frac{\frac{1}{6}C_0 + \frac{1}{2}C_1 + \frac{1}{2}C_0 + \frac{1}{2}C_1}{4}$

$$= \frac{1}{6}C_0 + \frac{1}{4}C_1$$

Thus

$$y = \sum_{n=0}^{\infty} C_n x^n$$

$$= C_0 + C_1 x + C_2 x^2 + C_3 x^3 + C_4 x^4 + \cdots$$

$$= C_0\left[1 + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{6}x^4 + \cdots\right] +$$

$$C_1\left[x + \frac{1}{2}x^2 + \frac{1}{2}x + \frac{1}{4}x^4 + \cdots\right]$$

② Substituting $y = \sum_{n=0}^{\infty} C_n x^n$ into the equation gives

$$(x-1)\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} - x\sum_{n=1}^{\infty} nC_n x^{n-1} + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\Rightarrow \sum_{n=2}^{\infty} n(n-1)C_n x^{n-1} - \sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} - \sum_{n=1}^{\infty} nC_n x^n + \sum_{n=0}^{\infty} C_n x^n = 0$$

$$\Rightarrow \sum_{n=2}^{\infty} n(n-1)C_n x^{n-1} - 2C_2 - \sum_{n=3}^{\infty} n(n-1)C_n x^{n-2} - \sum_{n=1}^{\infty} nC_n x^n + C_0 + \sum_{n=1}^{\infty} C_n x^n = 0$$

$$\Rightarrow -2C_2 + C_0 + \sum_{n=1}^{\infty} (n+1)nC_{n+1} x^n - \sum_{n=1}^{\infty} (n+2)(n+1)C_{n+2} x^n - \sum_{n=1}^{\infty} nC_n x^n + \sum_{n=1}^{\infty} C_n x^n = 0$$

Comparing coefficients

x^0 : $-2C_2 + C_0 = 0 \Rightarrow C_2 = \frac{1}{2}C_0$

x^n : $(n+1)nC_{n+1} - (n+2)(n+1)C_{n+2} - (n-1)C_n = 0$

$$\Rightarrow C_{n+2} = \frac{(n+1)nC_{n+1} - (n-1)C_n}{(n+2)(n+1)}, \quad n = 1, 2, 3, \ldots$$

For $n=1$; $C_3 = \frac{2C_2}{3\cdot 2} = \frac{C_0}{3\cdot 2}$

For $n=2$; $C_4 = \frac{3(2)C_3 - C_2}{4\cdot 3} = \frac{C_0 - \frac{1}{2}C_0}{4\cdot 3}$

$$= \frac{C_0}{4\cdot 3\cdot 2}$$

$y = C_0 + C_1 x + C_2 x^2 + C_3 x^3 + C_4 x^4 + \ldots$

$= C_0 + C_1 x + \frac{1}{2}C_0 x^2 + \frac{C_0}{3!}x^3 + \frac{C_0}{4!}x^4 + \ldots$

$\Rightarrow y' = C_1 + C_0 x + \frac{C_0}{2!}x^2 + \frac{C_0}{3!}x^3 + \ldots$

$y(0) = C_0 = -2$

$y'(0) = C_1 = 6$

$$\therefore y = -2 + 6x - \frac{2}{2!}x^2 - \frac{2}{3!}x^3 - \frac{2}{4!}x^4 - \cdots$$
$$= -2\left[1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots\right] + 8x$$
$$= -2e^x + 8x$$

③ Substituting $y = \sum_{n=0}^{\infty} C_n x^n$ gives

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + \sin(x)\sum_{n=0}^{\infty} C_n x^n = 0$$

$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$, so

$$\sum_{n=2}^{\infty} n(n-1)C_n x^{n-2} + \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots\right)\sum_{n=0}^{\infty} C_n x^n = 0$$

$$\Rightarrow \left[2C_2 + 3\cdot 2C_3 x + 4\cdot 3C_4 x^2 + 5\cdot 4C_5 x^3 + 6\cdot 5C_6 x^4 + \cdots\right] +$$
$$\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots\right)\left(C_0 + C_1 x + C_2 x^2 + C_3 x^3 + C_4 x^4 + \cdots\right) = 0$$

$$\Rightarrow \left[2C_2 + 6C_3 x + 12C_4 x^2 + 20C_5 x^3 + 30C_6 x^4 + \cdots\right] +$$
$$\left(C_0 x + C_1 x^2 + \left(C_2 - \tfrac{1}{3!}C_0\right)x^3 + \left(C_3 - \tfrac{1}{3!}C_1\right)x^4 + \cdots\right) = 0$$

$$\Rightarrow 2C_2 + (C_0 + 6C_3)x + (C_1 + 12C_4)x^2 + \left(C_2 - \tfrac{1}{3!}C_0 + 20C_5\right)x^3$$
$$+ \left(C_3 + 30C_6 - \tfrac{1}{3!}C_1\right)x^4 + \cdots = 0$$

Comparing coefficients

x^0: $2C_2 = 0 \Rightarrow C_2 = 0$

x^1: $C_0 + 6C_3 = 0 \Rightarrow C_3 = -\frac{1}{6}C_0$

x^2: $C_1 + 12C_4 = 0 \Rightarrow C_4 = -\frac{1}{12}C_1$

x^3: $C_2 - \frac{1}{3!}C_0 + 20C_5 = 0 \Rightarrow C_5 = \frac{1}{20}\left(\frac{1}{6}C_0 - C_2\right) = \frac{1}{120}C_0$

x^4: $C_3 + 30C_6 - \frac{1}{6}C_1 = 0 \Rightarrow C_6 = \frac{1}{30}\left(\frac{1}{6}C_1 - C_3\right)$
$= \frac{1}{30}\left(\frac{1}{6}C_1 + \frac{1}{6}C_0\right)$
$= \frac{1}{180}C_1 + \frac{1}{180}C_0$

$$\therefore y = C_0\left[1 - \frac{1}{6}x^3 + \frac{1}{120}x^5 + \frac{1}{180}x^6 + \cdots\right] +$$
$$C_1\left[x - \frac{1}{12}x^4 + \frac{1}{180}x^6 + \cdots\right]$$